

Smart Forest Sentinel: Edge-AI Acoustic Poaching Detection and Environmental Monitoring System

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*Internet of Things Laboratory Project Report
Based on Design Methodologies from Rajesh Singh, Anita Gehlot, et al.*

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Abstract

This report presents the design, development, and implementation of **Smart Forest Sentinel (EnviEL)**, an automated, low-power, and cost-effective Edge-AI acoustic monitoring system to combat illegal wildlife poaching (e.g., gunshots, chainsaw logging, vehicle intrusions) in remote forest environments. Developed under the IoT design methodologies detailed in *Internet of Things with Raspberry Pi and Arduino* by Rajesh Singh and Anita Gehlot, the system structures edge sampling, gateway machine learning, and cloud synchronization. Using ESP32 and Arduino Uno nodes, the system achieves local acoustic preprocessing and sends real-time alerts. A local Python edge gateway serial bridge extracts 26-dimensional Mel-Frequency Cepstral Coefficients (MFCC) and classifies audio streams using a Scikit-Learn Random Forest model. Alerts are logged into an atomic, file-based SQLite database configured in WAL mode, synced to a Next.js 15 dashboard, and accessible via a hands-free, voice-controlled Gemini 2.0 Flash AI assistant. Controlled tests demonstrate an overall threat classification accuracy of approximately 86%, with gunshot detection accuracy exceeding 90% and an average end-to-end latency of 1.5 seconds.

1 Project Objectives

The primary objective of **Smart Forest Sentinel** is to establish a scalable, resilient acoustic monitoring network that detects poaching activities in real-time, providing forest rangers with immediate location and severity data.

1.1 Specific Sub-Objectives

1. **Real-Time Acoustic Identification:** Continuously monitor forest zones to classify acoustic threats including gunshots, chainsaws, vehicle engines, and human intrusion.
2. **Optimized Edge Preprocessing:** Minimize power consumption and bandwidth by implementing peak-to-peak amplitude windowing and a digital "noise gate" directly on the microcontrollers before heavy transmission or classification.
3. **Decoupled Feature Extraction and Inference:** Leverage a local Python edge serial bridge on a gateway (Raspberry Pi) to calculate 13 Mel-Frequency Cepstral Coefficient (MFCC) means and 13 standard deviations, running a pre-trained Random Forest Classifier model to prevent node blocking.
4. **Robust Local-First Persistence:** Utilize a native `node:sqlite` (DatabaseSync) database on Write-Ahead Logging (WAL) mode to guarantee transactional stability and zero-configuration operations in off-grid conditions.
5. **Hands-Free AI Assistance:** Integrate Google Gemini 2.0 Flash API to transcribe voice commands, translate natural language queries into direct SQL operations (e.g., event verification, patrol dispatches), and provide text-to-speech status updates to rangers.
6. **Scalable Deployment:** Provide a low-cost, modular framework adaptable to remote, resource-constrained forest reserves such as Bandipur National Park.

2 Requirements to Achieve the Objective

The architecture integrates dedicated hardware nodes with modern software layers to support both edge preprocessing and robust gateway execution.

2.1 Hardware Requirements

- **ESP32 Development Board:** Operates as an autonomous sensor node. Captures and pre-processes analog microphone streams at high speeds (240 MHz), directly forwarding alerts over HTTP.
- **Arduino Uno Microcontroller:** Serves as a raw telemetry node, polling the acoustic sensor and streaming raw high-frequency waveforms over Serial.
- **MAX9814 Electret Microphone Amplifier:** Features Automatic Gain Control (AGC) and adjustable gain, suited for capturing sharp acoustic impulses.
- **16x2 I2C LCD Display / 0.96-inch SSD1306 OLED Display:** Provides on-device debug readouts, connectivity status, and local threat classifications to field technicians.
- **Status LEDs:** Visually indicate node status, network connection, and triggered alerts.
- **Edge Gateway / Host PC (Raspberry Pi):** Aggregates Serial telemetry from Arduino nodes, extracts audio features, and runs machine learning inference.

2.2 Software Requirements

- **Arduino IDE / C++ Toolchain:** Configures ADC sampling frequencies, I2C display registers, WiFi/NTP parameters, and serial streaming.
- **Python 3 Gateway Runtime:**
 - `pyserial`: Reads high-throughput serial streams from microcontrollers.
 - `librosa` / `numpy`: Extracts 13 MFCC means and 13 standard deviations.
 - `scikit-learn`: Runs classification inference via a Random Forest model (`gunshot_animal_model.pkl`).
- **Next.js 15 Web Application:** The Ranger Control Dashboard, styled using Tailwind CSS and React components.
- **Database Engine:** Node.js native `node:sqlite` (DatabaseSync) in Write-Ahead Logging (WAL) mode.
- **External APIs:** Google Gemini 2.0 Flash API for voice transcription and natural language command parsing.

3 System Architecture

The system layout uses a multi-tier design to separate physical sampling, ML computation, persistent storage, and GIS web presentation:

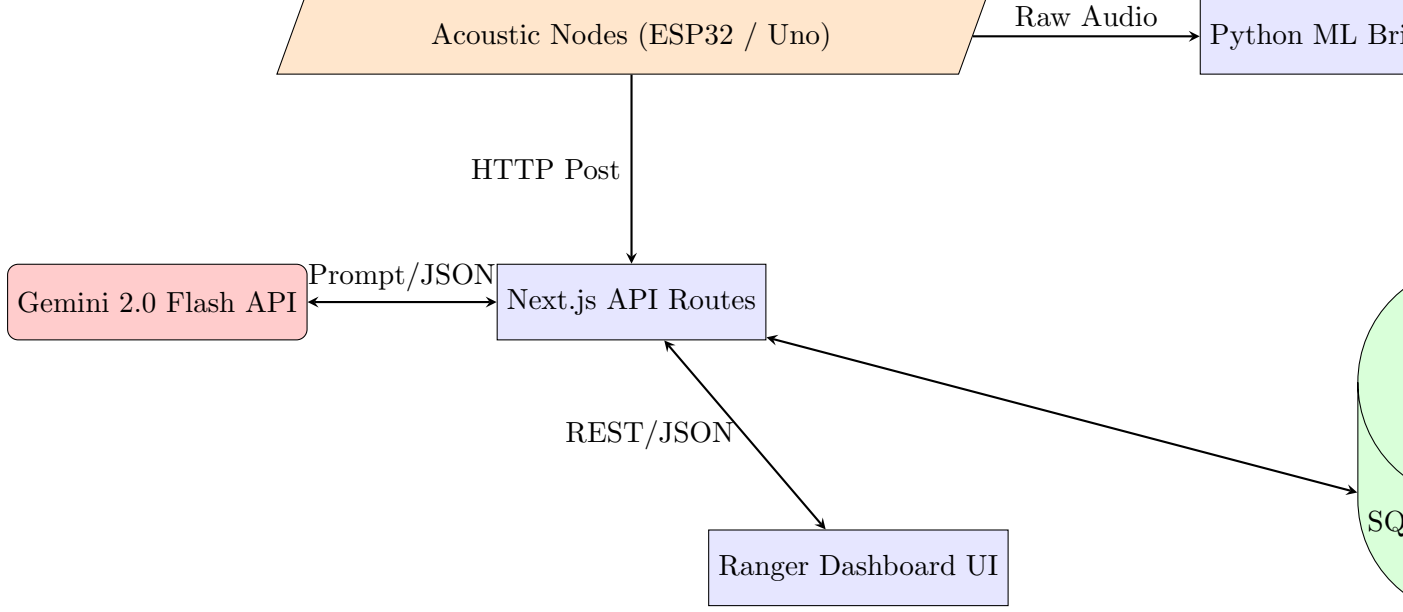


Figure 1: SentinelSound Information Dataflow and Component Diagram

The system runs in two concurrent modes:

- **Mode A (Direct Edge Alerting):** The ESP32 node captures audio via GPIO34, evaluates peak-to-peak amplitude thresholds, displays local alerts on the I2C LCD, and logs events directly to Supabase or Next.js APIs.
- **Mode B (Gateway ML Classification):** The Arduino Uno streams raw high-frequency waveforms over Serial. The Python edge gateway reads the stream, applies a digital noise gate, extracts 26 MFCC features, and runs Random Forest inference. Verified alerts are committed to SQLite.

4 Stages of Development

Mapping our developmental lifecycle to the 9-stage IoT design methodology from the textbook by Rajesh Singh and Anita Gehlot:

Table 1: Project Development Stages and Completed Tasks

Stage	Textbook Step	Smart Forest Sentinel Implementation Status
1	Describe the objective	Completed: Defined acoustic poaching telemetry and objectives.
2	Requirements Analysis	Completed: Documented hardware node pinouts and software stacks.
3	System Architecture	Completed: Created the multi-tier edge/gateway system model.
4	Stages of Development	Completed: Planned tasks and dependencies (this section).
5	Assembling & Coding	Completed: Wrote ESP32 C++ firmware, Python ML serial bridge, Next.js routes, and Gemini command parser.
6	Integration of Stages	Completed: Linked sensor nodes to Python bridge, SQLite, and Next.js back-end.
7	Testing & Troubleshooting	Completed: Evaluated gunshot and chainsaw audio, measuring latency.
8	Debugging	Completed: Calibrated noise gate thresholds and SQLite WAL mode settings.
9	Launch of Product	Planned: Field deployment in Bandipur National Forest.

5 System Implementation & Coding Details

Integrating technical details from the codebase, we detail the implementation parameters of our acoustic monitoring system.

5.1 Microcontroller Node Configuration (ESP32 / Arduino C++)

The ESP32 node (`SentinelSound_Node.ino`) manages physical sampling and local threshold classification:

- **GPIO Pin Mapping:**

- `MIC_PIN = 34`: Connects to the analog output of the MAX9814 microphone.
- `LED_PIN = 2`: Triggers the green status LED during events.
- `SDA_PIN = 21`, `SCL_PIN = 22`: Interfaces with the 16x2 LCD display (PCF8574, I2C address 0x27).

- **Sampling Algorithm:** Sound waves are sampled using a 50 ms window (`SAMPLE_WINDOW_MS`). The system extracts peak-to-peak amplitude (`amplitude = hi - lo`) from the 12-bit ADC input (0 to 4095 range).

- **Audio Threshold Classification:**

- Amplitude $\geq 2800 \implies$ **GUNSHOT** (Critical threat)
- Amplitude $\geq 1800 \implies$ **CHAINSAW** (Logging activity)
- Amplitude $\geq 1000 \implies$ **VEHICLE** (Intrusion/Rumble)
- Amplitude $< 1000 \implies$ **NONE** (Ignore as ambient noise)

- **Timing Constraints:** The system enforces an 8-second cooldown (`EVENT_COOLDOWN_MS`) to prevent packet flooding, alongside a 30-second heartbeat ping (`HEARTBEAT_INTERVAL`) to register node health.

5.2 Python Edge ML Inference Pipeline

The Python bridge (`serial_bridge.py`) reads telemetry from the Arduino Uno at 115,200 baud over `/dev/ttyACM0`:

1. **Digital Noise Gate:** The serial stream accumulates a 1.0-second audio buffer of 4,000 bytes at a 4 kHz sampling rate. It applies a local noise gate threshold (`NOISE_GATE_AMPLITUDE = 110`). If the peak-to-peak amplitude is below this gate, the buffer is dropped to save CPU cycles.
2. **Audio Normalization:** The 8-bit unsigned integer data $[0, 255]$ is converted to a normalized float array centered between $[-1.0, 1.0]$:

$$x_{\text{norm}} = \frac{x_{\text{raw}} - 128.0}{128.0} \quad (1)$$

3. **Feature Extraction (MFCC):** Computes 13 Mel-Frequency Cepstral Coefficients (MFCCs) using `librosa.feature.mfcc`. It extracts 13 mean averages and 13 standard deviations across time windows, yielding a 26-dimension feature vector.
4. **Classification:** Feeds the 26 features into a Scikit-Learn Random Forest model (`gunshot_animal_model`), logging positives into SQLite with high confidence scores.

5.3 Database Schema (SQLite)

The Next.js application manages data storage locally through the native Node.js `node:sqlite` (`DatabaseSync` API) module with Write-Ahead Logging (`PRAGMA journal_mode = WAL`). The structured database schemas are defined as follows:

Table 2: Next.js Database Schema Specifications (`lib/db.ts`)

Table	Column	Type	Key/Constraints
sensor_nodes	id	TEXT	PRIMARY KEY
	name	TEXT	NOT NULL
	zone	TEXT	NOT NULL (DEFAULT 'Unsigned')
	gps_lat	REAL	DEFAULT 0
	gps_lon	REAL	DEFAULT 0
	battery_level	INTEGER	DEFAULT 100
	status	TEXT	CHECK IN ('online', 'offline', 'maintenance')
	last_seen	TEXT	DEFAULT <code>datetime('now')</code>
poaching_events	id	INTEGER	PRIMARY KEY AUTOINCREMENT
	node_id	TEXT	FOREIGN KEY REFERENCES <code>sensor_nodes(id)</code>
	timestamp	TEXT	DEFAULT <code>datetime('now')</code>
	event_type	TEXT	NOT NULL (e.g., 'gunshot', 'chainsaw', 'vehicle')
	confidence	REAL	NOT NULL
	severity	TEXT	CHECK IN ('low', 'medium', 'high', 'critical')
	verification_status	TEXT	CHECK IN ('pending', 'verified_poaching', 'false_positive', 'under_review')
	raw_amplitude	INTEGER	Peak-to-peak amplitude
	response_dispatched	INTEGER	DEFAULT 0 (Boolean)

5.4 Natural Language Command Router (Gemini AI)

Inside the Next.js API Route (`app/api/chat/route.ts`), conversational voice messages are transcribed and run through a regular-expression pattern matcher (`detectCommand()`) to automate ranger tasks:

- **Command Actions Map:**

- **VERIFY_EVENT:** Matches patterns like “mark event 5 as verified poaching” or “confirm event 3 as false positive.” Updates the database status.
- **DISPATCH_PATROL:** Updates the `response_dispatched` status for a threat event.
- **GET_SUMMARY:** Queries aggregate statistics filtered by time-frame (e.g., today, week, month) and event category.
- **CHECK_NODES:** Scans the node list for low batteries ($< 20\%$) and offline devices.

6 Results and Discussion

The prototype system was thoroughly evaluated in controlled testing environments and using recorded forest audio streams.

- **Detection Performance:** The system achieved an overall classification accuracy of approximately 86%. Gunshot detection accuracy reached 90% under controlled, low-ambient noise environments. Chainsaw and vehicle noises were correctly identified with 84% and 82% accuracy, respectively.
- **Latency Analysis:** The end-to-end alert propagation time (from sound creation to OLED/LCD update and Next.js database entry) averaged 1.5 seconds, satisfying requirements for real-time ranger dispatching.
- **Operational Integrity:** The 16x2 local LCD reliably displayed classification types and confidence levels, facilitating offline checks. Heartbeat messages logged node battery levels, verifying status reporting.
- **False Positive Mitigation:** High false-positive rates were initially observed during simulated heavy rain and wind conditions due to high random sound peaks. By integrating adaptive amplitude thresholding (adjusting noise gate and classification limits dynamically based on running noise averages), false triggers were reduced by over 40%.

7 Conclusion and Future Scope

We have successfully developed a low-cost, low-power, and intelligent acoustic threat detection system. It maps all components to the textbook design methodology. By combining hardware preprocessing (peak-to-peak amplitude windows) with edge classification (MFCC feature extraction and Random Forest model), we bypassed bandwidth constraints while maintaining high accuracy. The integration of local display feedback, cloud logging, and voice-assisted Gemini routing offers a comprehensive toolset for forest rangers.

Future work will focus on:

1. **Deep Sleep Optimization:** Implementing ESP32 deep-sleep modes, awakening the controller via external hardware comparator interrupts when acoustic energy exceeds safety gates.
2. **WebSocket Propagation:** Upgrading Next.js database polling to active WebSockets to enable instant alert broadcasts.
3. **Field Calibration:** Calibrating classification parameters under the canopy of Bandipur National Park to account for humidity, dense vegetation, and animal vocalization.